MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

1/ TYPE DOD-026

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Planar triode
See figure 1
Mounting position: Any
Weight: 0.16 ounce nominal (4.725 grams)

ABSOLUTE RATINGS:

Parameter:	F1	Ef	Eb	Ec	Ib	Ic	Ik	Pp	tk	Cooling	T(seal)
Unit:	GHz	V	Vdc	Vdc	mAdc	mAdc	mAdc	W	sec	---	°C
Maximum:	---	6.6	400	-15	15	3	15	2.25	---	---	150
Minimum:	---	6.0	---	---	---	---	---	---	60	Conduction	---
			(Note 6)								
TEST CONDITIONS:	5.4	6.3	175	Adj	10	---	---	---	180	Conduction	---

GENERAL:

Qualification - Required

1/ This is a selected tube (JAN type 7391) for the special requirements of the AN/APN-141 application.

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1211	Insulation of electrodes	Fig 2	Ebb = -500 Vdc; Ekk = +45 Vdc	0.65 0.65	II II	Rgp Rgk	5.0 0.25	--- ---	Meg Meg
1231	Pulsing emission	1	Ef = 5.0 V; Ec = -18.0 Vdc; eb = 390 v (peak-to-peak); ec = 46 v (peak-to-peak); Rg = 470 ohms; Rk = 10 ohms	0.65	II	Is	3.5	---	mAdc
1301	Heater current	-		0.65	II	If	360	400	mA
1261	Electrode voltage (1) (grid)	-		0.65	II	Ec	-0.70	-2.55	Vdc
1306	Transconductance	-		0.65	II	Sm	8,000	13,500	μmhos
1236	Power oscillation	2	Eb = 150 Vdc; Ib = 15 mAdc; Rg = 2,000 ohms; F = 5.4 GHz (min)	0.65	II	Po	30	---	mW
1261	Electrode voltage (2) (grid)	-	Eb = 400 Vdc; -Ec/Ib = 50 μAdc	0.65	II	Eco	---	-13	Vdc
	Quality conformance inspection, part 2								
1316	Amplification factor	-		---	---	Mu	46	80	---
1331	Direct-interelectrode capacitance	-	No voltages; Drawing 270-JAN	---	---	Cin Cgp Cout	2.60 1.40 ---	3.95 1.80 0.023	pF pF pF
2126	Envelope strain	3		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	-	Group B; power oscillation; t = 500 hours; DESC Drawing No. D66002	---	---	---	---	---	---
---	Life-test end point:								
1236	Power oscillation	-		---	---	ΔPo t	---	-20	%
	Periodic-check tests								
1031	Low-frequency vibration	4	Eb/Ib = 10 mAdc; Ec = 0; Rp = 10,000 ohms; Drawing 273-JAN	---	---	Ep	---	100	mVac
1041	Shock, specified pulse	5	No voltages; 300 G; Drawing 274-JAN	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Periodic-check tests -Continued								
---	Shock test end points:								
1306	Transconductance	-		---	---	Sm	8,000	13,500	μmhos
1211	Insulation of electrodes	-		---	---	Rgp	5.0	---	Meg
1211	Insulation of electrodes	-		---	---	Rgk	0.25	---	Meg

NOTES:

1. The 60-cycle voltages eb and ec must be in phase, and the sinusoidal waveform should not be distorted. Read average cathode current.
2. Operate in cavity, DESC Drawing No. D66001. Adjust tuning for maximum nonsquegging power output.
3. Envelope strain procedure. Tubes shall be tested as specified in method 2126, except that they shall first be immersed in water at not more than 5°C for 15 seconds, and immediately thereafter subjected to the standard temperature cycle specified in method 2126. The criteria for air leaks shall be heater current of 410 mA, or more.
4. Vibrate in plane perpendicular to the plane of the grid. Test 10 tubes selected at random from the first production lot of each calendar year, and approximately every 90 days of production during the year. If more than one tube fails to pass the specified end points, the test shall become a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3 on all lots in process. After three consecutive successful submissions, the test shall revert back to the 10-tube quarterly test.
5. Apply the force perpendicular to the plane of the grid from the cathode end. Test 10 tubes selected at random from the first production lot of each calendar year. If more than one tube fails to pass the specified end points, the test shall become a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3 on all lots in process. After three consecutive successful submissions, the test shall revert back to a 10-tube annual test.
6. Maximum anode supply voltage in pulse service shall be 400 Vdc. Maximum value for CW service shall be 200 Vdc.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

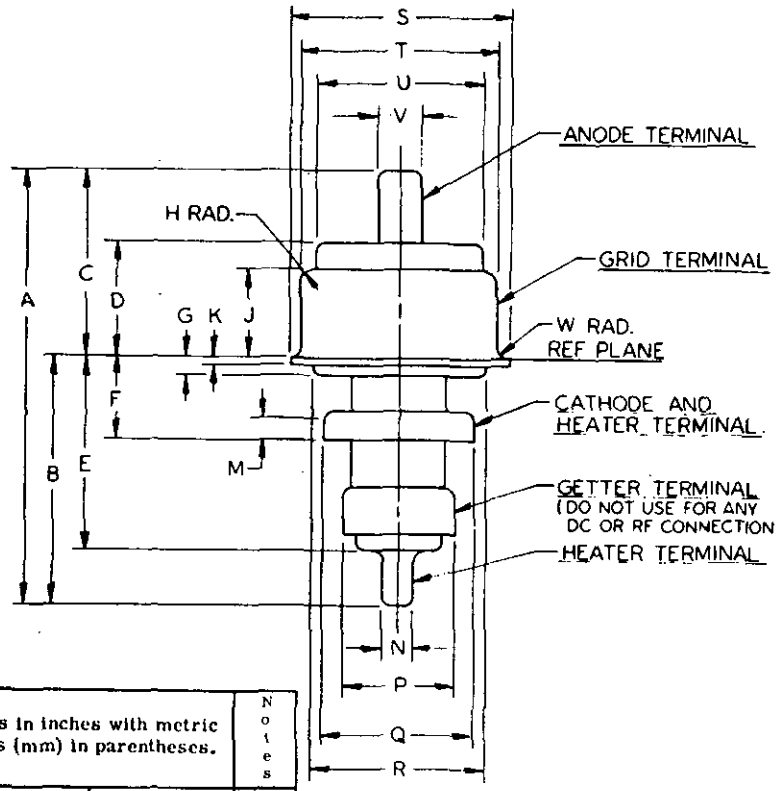
User activities:
Army - MU
Navy - AS, OS, MC, CG

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2765)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses.		Notes
	Minimum	Maximum	
Quality conformance inspection, part 2			b
A	.960 (24.38)	1.040 (26.42)	
B	.530 (13.46)	.590 (14.99)	
C	.410 (10.41)	.470 (11.94)	
D		.272 (6.91)	
E		.475 (12.07)	
F	.163 (4.14)	.193 (4.90)	
Quality conformance inspection, part 3 (periodic check)			a, b
G		.060 (1.52)	
H		.030 (.76)	
J	.190 (4.83)	.210 (5.33)	
K	.009 (.23)	.015 (.38)	
M	.040 (1.02)	.070 (1.78)	
N	.059 (1.50)	.065 (1.65)	
P		.257 (6.53)	
Q	.326 (8.28)	.334 (8.48)	
R		.385 (9.78)	
S	.483 (12.27)	.497 (12.62)	
T	.435 (11.05)	.445 (11.30)	e
U		.385 (9.78)	
V	.088 (2.24)	.094 (2.39)	f
W		.008 (.20)	

NOTES:

a. Test these dimensions on 10 tubes per month when in continuous production. Failure of more than one tube to meet the tolerance for any dimension shall cause only that dimension to become a 6.5 AQL, *S3* inspection level test. After 3 consecutive successful submissions, the test shall revert back to a 10-tube monthly test.

b. The AQL applies to each dimension separately, and is not combined.

c. All tubes are to be 100 percent tested for *concentricity in gage (see figure 3)*, *Tube shall* be rotated 360° in gage without binding.

d. Grid shell seal shall be completely filled to cathode ceramic shelf.

e. 'T' dimension shall be measured other than across tube designation etch, and approximately .100 (2.54 mm) from the reference plane.

f. 'V' dimension shall measured approximately .100 (2.54 mm) from the end of the anode terminal.

FIGURE 1. Outline drawing of electron tube type DOD-026.

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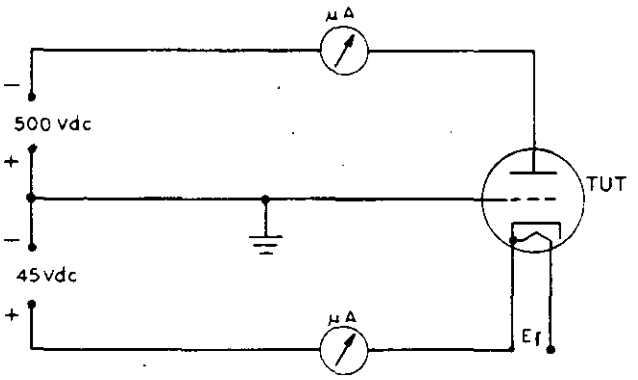
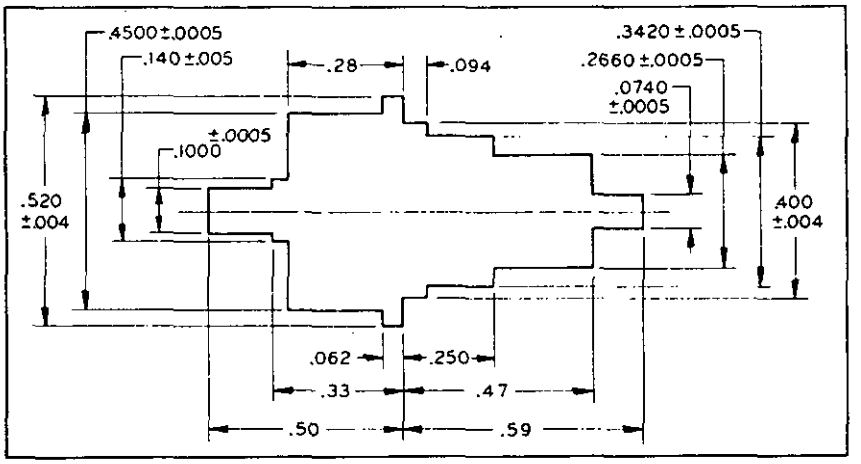


FIGURE 2. Insulation of electrodes test.



NOTE:
Unless otherwise specified, tolerance for dimension .250 or less ±.008; over .25 ±.02.

FIGURE 3. Gage.

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